AP BIOLOGY · UNIT 1

TOPIC 1.7

Proteins: Structure & Function

From amino acid chains to molecular machines — understanding the four levels of protein architecture and how structure determines function.

Learning Objectives & Roadmap

By the end of this review, you should be able to **describe the structure and function of proteins** at every level of organization — from individual amino acids to complex quaternary assemblies. Use this roadmap to track your progress through the essential knowledge statements.

01

Amino Acid Anatomy

Central carbon, R groups, and chemical properties

02

Peptide Bond Formation

Dehydration synthesis linking amino acids into polypeptides

03

Primary & Secondary Structure

Sequence → local folding (α-helices, β-sheets)

04

Tertiary & Quaternary Structure

3-D shape, interactions, and multi-subunit proteins

05

Structure → Function

How all four levels determine what a protein does

ESSENTIAL KNOWLEDGE 1.7.A

Anatomy of an Amino Acid

Every protein in your body — from hemoglobin to antibodies — is built from the same set of **20 common amino acids**. Despite their incredible diversity of function, all amino acids share an identical core architecture. Understanding this core structure is your foundation for everything else in this unit.

Central (Alpha) Carbon

The alpha carbon (Cα) sits at the center of every amino acid and serves as the attachment point for **four different groups**. This tetrahedral arrangement gives amino acids their characteristic geometry. Think of the alpha carbon as the "hub" of a wheel with four spokes.

Hydrogen Atom

A single hydrogen atom is bonded to the central carbon. While it might seem insignificant, this hydrogen helps establish the tetrahedral shape around the alpha carbon and contributes to the **chirality** (handedness) of the molecule — all biological amino acids are L-isomers.

Carboxyl Group (–COOH)

This acidic functional group can **donate a proton (H^+)**, giving amino acids part of their "acid" identity. At physiological pH (~7.4), the carboxyl group is typically deprotonated ($–COO^-$). It participates directly in **peptide bond formation** during protein synthesis.

Amine Group ($–NH_2$)

This basic functional group can **accept a proton**, making amino acids amphoteric (they can act as both acids and bases). At physiological pH, the amine group is usually protonated ($–NH_3^+$). It forms the other half of the **peptide bond** with the carboxyl group of the adjacent amino acid.

AP Exam Tip: Remember the mnemonic **"CHAN"** — Central carbon, Hydrogen, Amine group, and the Nth component: the variable R group (covered next). The College Board expects you to identify all four groups attached to the alpha carbon.

R GROUPS

The R Group: What Makes Each Amino Acid Unique

While the alpha carbon, hydrogen, amine group, and carboxyl group are **identical** across all 20 amino acids, the **R group (side chain)** is the variable component that gives each amino acid its distinct chemical personality. The R group is covalently bonded to the central carbon and can range from a single hydrogen atom (glycine) to complex ring structures (tryptophan). The chemical properties of R groups are what ultimately drive protein folding and function.

Hydrophobic / Nonpolar

These R groups consist mainly of **carbon and hydrogen atoms** arranged in chains or rings. They do not interact favorably with water and tend to cluster together in the **interior of proteins**, away from the aqueous environment. This "hydrophobic effect" is one of the strongest driving forces behind protein folding.

Examples: Leucine, Isoleucine, Valine, Phenylalanine, Alanine

Hydrophilic / Polar

Polar R groups contain atoms like **oxygen, nitrogen, or sulfur** that create partial charges, allowing them to form **hydrogen bonds with water**. These amino acids are typically found on the **surface of proteins**, where they interact with the surrounding aqueous solution and contribute to protein solubility.

Examples: Serine, Threonine, Asparagine, Glutamine, Tyrosine

Ionic / Charged

Some R groups carry a **full positive or negative charge** at physiological pH. Positively charged (basic) R groups include lysine, arginine, and histidine. Negatively charged (acidic) R groups include aspartate and glutamate. These charged groups form **ionic bonds (salt bridges)** that stabilize protein structure and participate in enzyme active sites.

Examples: Lysine (+), Glutamate (−), Aspartate (−), Arginine (+)

> **Key concept:** The interactions of these R groups determine the structure and function of that region of the protein. Hydrophobic R groups drive folding inward; polar and charged R groups face outward or form specific bonds that stabilize shape.

PEPTIDE BONDS

Peptide Bond Formation: Building the Chain

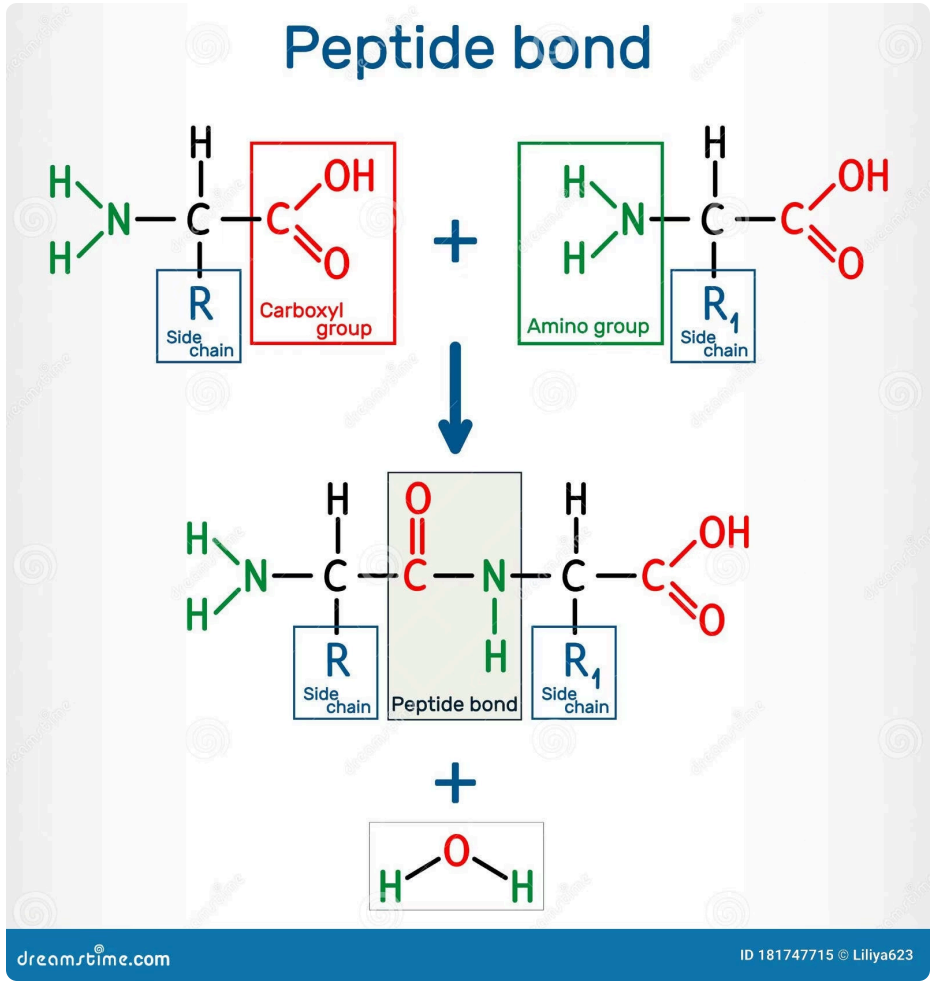
Proteins are built one amino acid at a time through a specific type of **dehydration synthesis reaction** that creates a **covalent peptide bond**. This is the fundamental chemical reaction that links amino acids into polypeptide chains — and it's the starting point of every protein in every living organism.

How It Works

The peptide bond forms between the **carboxyl group (–COOH)** of one amino acid and the **amine group ($-NH_2$)** of the next amino acid. During this condensation reaction:

1. The –OH from the carboxyl group and one –H from the amine group are removed
2. A molecule of **water (H_2O)** is released as a byproduct
3. A strong **covalent bond (C–N)** forms between the two amino acids
4. The resulting structure is called a **dipeptide**; further additions create a **polypeptide**

The peptide bond is **planar and rigid** — it does not rotate freely. However, the bonds on either side of the peptide bond *can* rotate, giving the polypeptide backbone its flexibility and allowing it to fold into complex shapes.



Vocabulary Check:

- **Dipeptide** = 2 amino acids joined
- **Tripeptide** = 3 amino acids joined
- **Polypeptide** = many amino acids joined (typically 50+)
- **Protein** = one or more polypeptides folded into a functional shape

PRIMARY STRUCTURE

Primary Structure: The Amino Acid Sequence

The **primary structure** of a protein is simply the **linear sequence of amino acids** in the polypeptide chain, read from the N-terminus (amine end) to the C-terminus (carboxyl end). While this might sound straightforward, the primary structure is arguably the most important level because **it determines all subsequent levels of folding and, ultimately, protein function**.

Genetically Determined

The amino acid sequence is encoded directly by the **nucleotide sequence of DNA** (via mRNA). Each set of three nucleotides (a codon) specifies one amino acid. A change in even a **single nucleotide** can alter the amino acid sequence — as seen in sickle cell disease, where one glutamate → valine substitution in hemoglobin's β-chain causes the protein to misfold.

Sequence Dictates Shape

The identity and order of amino acids determine which R group interactions are possible. Hydrophobic residues near each other will drive inward folding; charged residues in proximity may form salt bridges or repel each other. This means the **primary structure is the blueprint** for the protein's three-dimensional architecture.

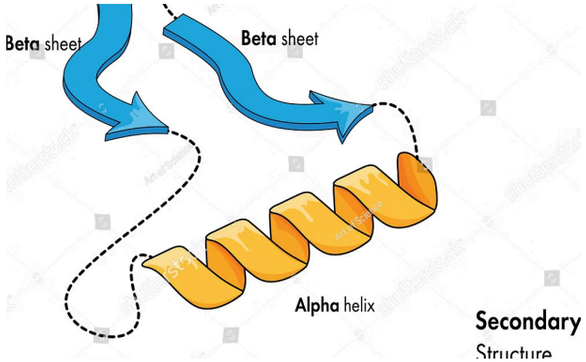
Held Together by Peptide Bonds

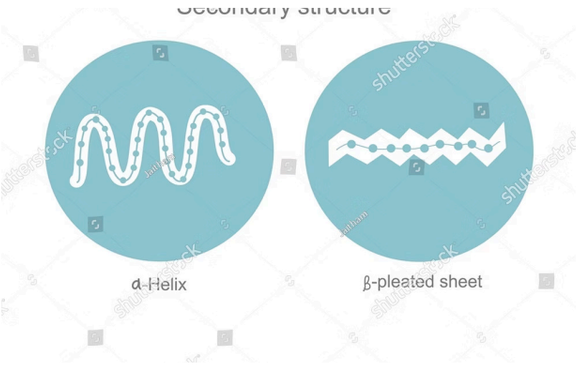
The peptide bonds holding the primary structure together are **strong covalent bonds** that require significant energy to break. This gives the polypeptide backbone its stability. However, the primary structure alone is not functional — the chain must fold into higher-order structures to perform its biological role.

SECONDARY STRUCTURE

Secondary Structure: Local Folding Patterns

Secondary structures emerge from **local folding** caused by **hydrogen bonds between atoms of the polypeptide backbone** — specifically between the N–H group of one amino acid and the C=O group of another. Crucially, these hydrogen bonds involve the backbone, **not the R groups**. Two major motifs dominate secondary structure:





Alpha Helix (α-helix)

The polypeptide backbone coils into a **right-handed spiral**, stabilized by hydrogen bonds between every 4th amino acid along the chain. The R groups project **outward from the helix** like bristles on a bottle brush. Alpha helices are common in structural proteins like **keratin** (hair, nails) and in transmembrane domains of membrane proteins, where hydrophobic R groups face the lipid bilayer.

Key H-bond pattern: C=O of residue *n* bonds to N–H of residue *n+4*

Beta Pleated Sheet (β-sheet)

Two or more segments of the polypeptide chain lie **side by side**, connected by hydrogen bonds between the backbone atoms of adjacent strands. The sheet has a characteristic **pleated (zigzag) appearance**, and R groups alternate above and below the plane of the sheet. β-sheets can run **parallel** (same direction) or **antiparallel** (opposite directions). They provide structural rigidity in proteins like **silk fibroin**.

Key H-bond pattern: C=O of one strand bonds to N–H of the adjacent strand

AP Exam Tip: The College Board frequently tests whether you can distinguish **backbone interactions** (secondary structure) from **R group interactions** (tertiary structure). If the question mentions hydrogen bonds between backbone atoms → secondary. If it mentions R group interactions → tertiary.

TERTIARY STRUCTURE

Tertiary Structure: The 3-D Shape

The tertiary structure is the **overall three-dimensional shape** of a single polypeptide chain. While secondary structure involves local folding of the backbone, tertiary structure results from **long-range interactions between R groups** that are often far apart in the primary sequence but brought close together when the chain folds. This is the level at which a protein becomes truly functional — enzymes gain their active sites, antibodies form their antigen-binding regions, and transport proteins create their binding pockets.

Four Types of R Group Interactions

Hydrogen Bonds

Form between **polar R groups**. A hydrogen atom bonded to an electronegative atom (O or N) is attracted to another electronegative atom on a nearby R group. These bonds are individually weak but collectively significant, especially on protein surfaces.

Hydrophobic Interactions

Nonpolar R groups are excluded from water and cluster together in the protein's interior. This "hydrophobic collapse" is often considered the **most important driving force** in protein folding, as it minimizes the disruption of water's hydrogen bonding network.

Ionic Interactions

Electrostatic attractions between **positively and negatively charged R groups** (e.g., lysine + and aspartate −). Also called **salt bridges**, these can be disrupted by changes in pH, which is why extreme pH denatures proteins.

Disulfide Bridges

Strong **covalent bonds** that form between the sulfur atoms of two **cysteine** residues (−S−S−). These are the strongest stabilizing force in tertiary structure and are especially important in proteins exposed to harsh environments (e.g., antibodies, extracellular proteins like keratin).

QUATERNARY STRUCTURE

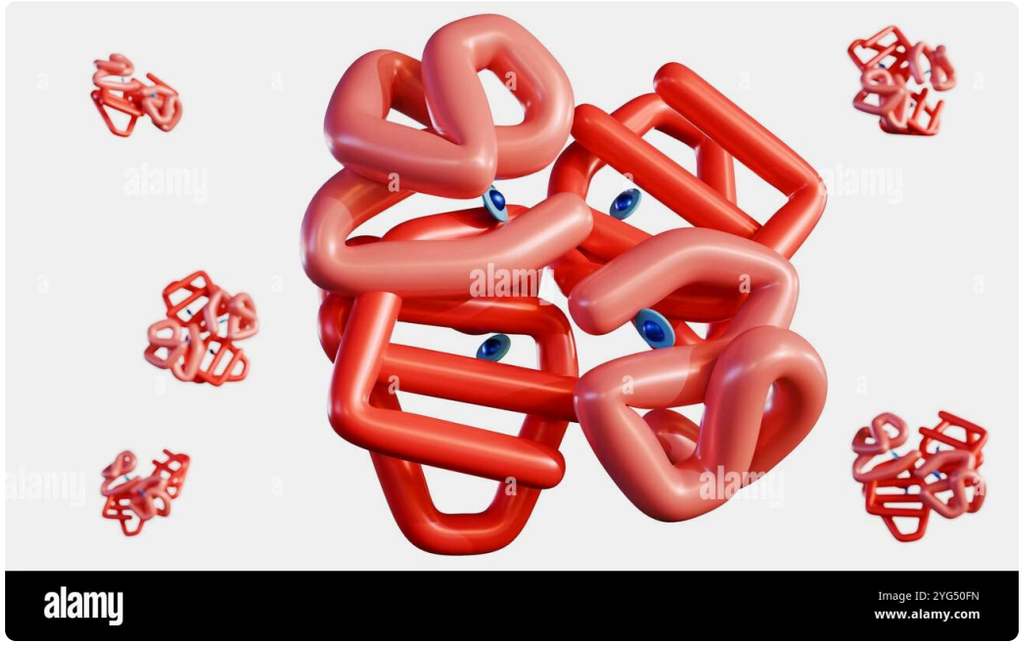
Quaternary Structure: Multi-Subunit Assembly

Not all proteins consist of a single polypeptide chain. The **quaternary structure** describes the arrangement and interactions of **two or more polypeptide subunits** that come together to form a functional protein complex. Each individual polypeptide is called a **subunit**, and the same types of interactions that stabilize tertiary structure (hydrogen bonds, hydrophobic interactions, ionic bonds, and sometimes disulfide bridges) hold the subunits together.

Classic Example: Hemoglobin

Hemoglobin is the oxygen-carrying protein in red blood cells, and it's the textbook example of quaternary structure. It consists of **four polypeptide subunits**: two α (alpha) chains and two β (beta) chains. Each subunit contains a **heme group** with an iron atom that binds one O_2 molecule, so one hemoglobin molecule can carry **up to four O_2 molecules**.

The subunits exhibit **cooperative binding** — when one subunit binds O_2, it changes shape slightly and makes it easier for the remaining subunits to bind O_2. This is a direct consequence of quaternary structure: the subunits physically interact and influence each other's conformation.

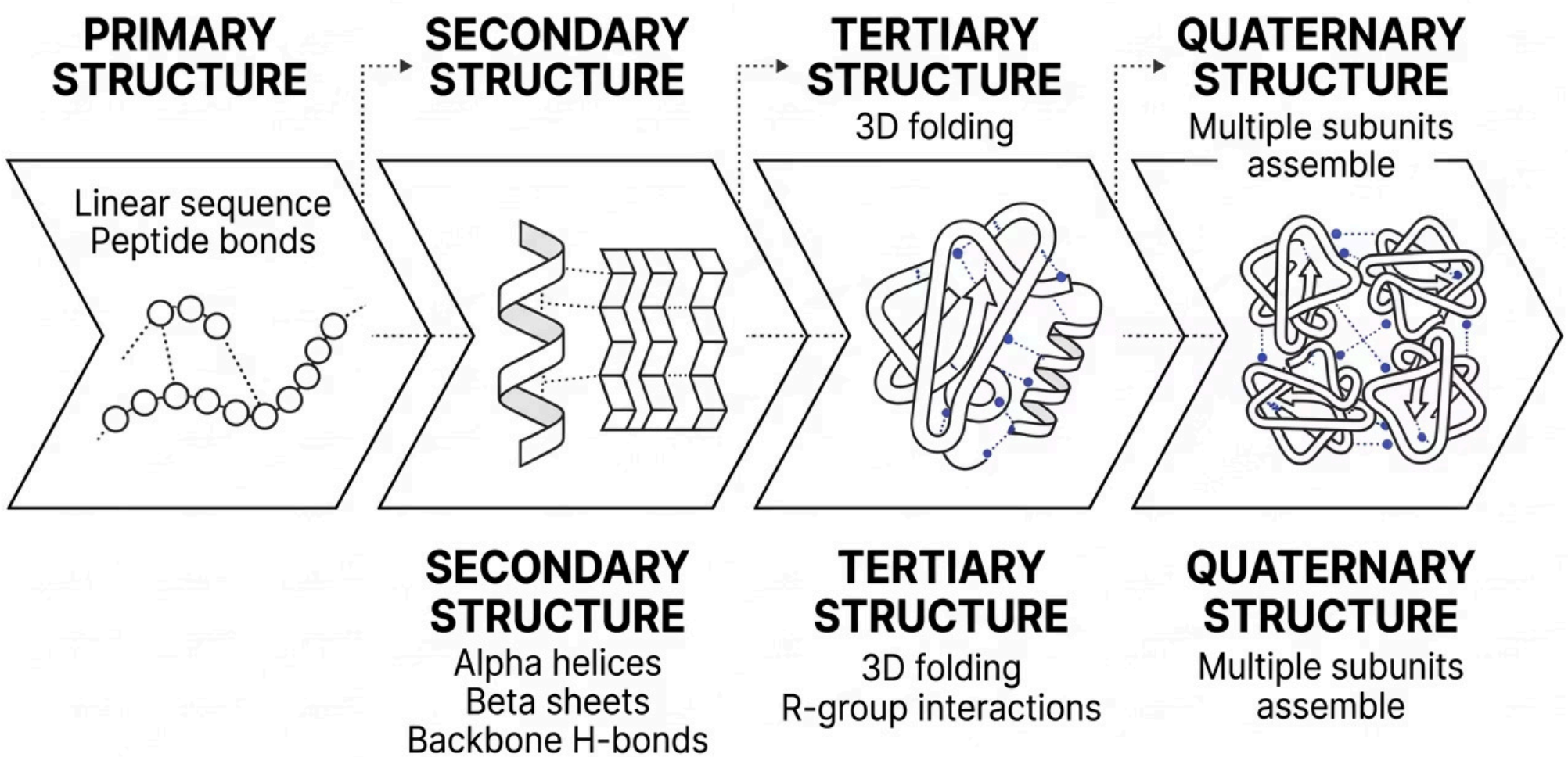


Hemoglobin's four subunits (2 alpha in one color, 2 beta in another) exemplify how quaternary structure enables cooperative function.

Other examples of quaternary proteins: Collagen (triple helix of 3 polypeptides), DNA polymerase (multiple subunits), antibodies (2 heavy + 2 light chains), ATP synthase (multiple subunits forming a rotary motor).

Four Levels at a Glance

Use this visual summary to review how each level of protein structure builds upon the previous one. Remember: **all four levels collectively determine the function of a protein**. If any level is disrupted (by mutation, pH change, temperature, etc.), the protein can lose its shape — a process called **denaturation** — and cease to function.



Each level emerges from specific chemical interactions. Primary structure uses **covalent peptide bonds**. Secondary structure uses **backbone hydrogen bonds**. Tertiary structure uses **R group interactions** (hydrogen bonds, hydrophobic interactions, ionic bonds, disulfide bridges). Quaternary structure uses **inter-subunit interactions** of the same types as tertiary.

BIG IDEA

Structure Determines Function

This is perhaps the single most important concept in all of AP Biology: **the shape of a biological molecule determines what it can do**. For proteins, this means that the precise arrangement of amino acids at every structural level creates the specific 3-D conformation that allows the protein to perform its unique biological role.

Enzymes

Enzymes have a precisely shaped **active site** — a pocket or groove formed by specific amino acid R groups. The substrate fits into this active site like a key into a lock (or more accurately, through an induced fit). If the tertiary structure is disrupted, the active site collapses and the enzyme cannot catalyze its reaction.

Antibodies

Antibodies have Y-shaped quaternary structures with **variable regions** at the tips that are uniquely shaped to bind one specific antigen. The incredible diversity of antibody shapes is generated by varying the amino acid sequences in these variable regions — a direct example of primary structure determining function.

Membrane Proteins

Transport proteins and receptors embedded in cell membranes have **hydrophobic α-helices** that span the lipid bilayer and hydrophilic regions that face the aqueous environment. This precise arrangement of polar and nonpolar R groups determines the protein's orientation and function within the membrane.

Structural Proteins

Proteins like **collagen** and **keratin** rely on repetitive secondary structures (triple helices, α-helices) to provide tensile strength and rigidity. Their elongated fibrous shapes — a direct result of their amino acid sequences and folding patterns — make them ideal for structural roles in skin, bone, hair, and connective tissue.

R Group Interactions: Quick Reference

This table summarizes the major types of interactions that contribute to protein structure at the tertiary and quaternary levels. Memorizing these distinctions is essential for the AP exam, as multiple-choice and free-response questions frequently ask you to identify which interaction is responsible for a given structural feature.

Interaction Type	R Groups Involved	Description	Strength & Notes
Hydrogen Bonds	Polar R groups	Attraction between a H bonded to O/N and a nearby electronegative atom	Weak individually; strong collectively. Disrupted by heat.
Hydrophobic Interactions	Nonpolar R groups	Nonpolar R groups cluster in the protein interior to avoid water	Major driving force of folding. Disrupted by detergents.
Ionic Bonds (Salt Bridges)	Charged R groups (+/−)	Electrostatic attraction between oppositely charged R groups	Moderate strength. Disrupted by pH changes.
Disulfide Bridges	Cysteine (−SH groups)	Covalent S−S bond between two cysteine residues	Strongest stabilizing force. Require reducing agents to break.

Remember: Backbone hydrogen bonds = **secondary structure**. R group interactions = **tertiary structure**. The same R group interactions also stabilize **quaternary structure** between subunits.

⚠ DENATURATION

What Happens When Structure Is Disrupted?

Since all four levels of protein structure determine function, any disruption to folding can have catastrophic consequences. **Denaturation** is the loss of a protein's three-dimensional shape due to disruption of the non-covalent interactions (and sometimes disulfide bridges) that maintain its higher-order structure. A denatured protein retains its primary structure (peptide bonds are intact) but loses its secondary, tertiary, and quaternary organization — and with it, its biological activity.

Causes of Denaturation

Heat disrupts hydrogen bonds and hydrophobic interactions. **Extreme pH** alters ionic charges on R groups, breaking salt bridges. **Detergents** disrupt hydrophobic interactions. **Heavy metals** (Pb^{2+}, Hg^{2+}) bind to sulfhydryl groups, breaking disulfide bridges. **Urea/organic solvents** disrupt hydrogen bonding networks.

Loss of Shape

The protein unfolds, losing its precise 3-D conformation. Active sites collapse, binding pockets disappear, and subunit interfaces separate. The polypeptide chain becomes a random coil — still a chain of amino acids, but without the organized structure needed to function.

Loss of Function

A denatured enzyme can no longer catalyze its reaction. A denatured receptor can no longer bind its ligand. A denatured structural protein can no longer provide support. In most cases, denaturation is **irreversible** (think: cooking an egg — you can't "uncook" it).

Frying an egg is a classic example of irreversible protein denaturation — heat disrupts the hydrogen bonds in albumin (egg white protein), causing it to change from transparent and liquid to opaque and solid.

REVIEW

Key Takeaways for the AP Exam

Here's your final study checklist for Topic 1.7. Make sure you can confidently explain each of these concepts — they appear frequently on both the multiple-choice and free-response sections of the AP Biology exam.

1 Amino acids have a universal core structure

Central carbon + H + amine group + carboxyl group + variable R group. The R group determines the chemical personality (nonpolar, polar, or charged/ionic) and, in turn, how that amino acid contributes to protein structure.

2 Peptide bonds link amino acids via dehydration synthesis

A covalent bond forms between the carboxyl group of one amino acid and the amine group of the next, releasing water. This creates a growing polypeptide chain.

3 Primary structure = amino acid sequence

Genetically determined and held by peptide bonds. Changes in sequence (mutations) can alter all higher levels of structure.

4 Secondary structure = local backbone folding

α-helices and β-pleated sheets stabilized by hydrogen bonds between backbone atoms (not R groups).

5 Tertiary structure = 3-D shape from R group interactions

Hydrogen bonds, hydrophobic interactions, ionic bonds, and disulfide bridges between R groups create the protein's functional conformation.

6 Quaternary structure = multi-subunit assembly

Two or more polypeptides interact via the same forces as tertiary structure to form a functional complex (e.g., hemoglobin).

7 All four levels determine function

Structure → function is a core AP Biology theme. Disruption of any level (denaturation) leads to loss of function.

PRACTICE

Test Yourself: Practice Questions

Use these questions to check your understanding before the exam. Try answering each one before reading the explanation.

1	**A single amino acid substitution in the primary structure of hemoglobin causes sickle cell disease. Explain how this change at one structural level affects the other levels and ultimately the protein's function.** **Think about:** Glutamate (charged, hydrophilic) → Valine (nonpolar, hydrophobic) in the β-chain. How does this change the R group interactions? What happens to tertiary and quaternary structure? Why do sickle-shaped red blood cells form?
2	**A student adds a strong acid to a solution containing an enzyme. The enzyme stops working. Has the primary structure changed? Explain.** **Think about:** What bonds does acid break? Are peptide bonds affected? What happens to R group charges at low pH? Can the enzyme refold if you neutralize the pH?
3	**Explain why a protein composed entirely of hydrophobic amino acids would not function properly in an aqueous cellular environment.** **Think about:** What would happen to the protein's solubility? Could it fold into a globular shape with a hydrophilic surface? How would this affect its interactions with other molecules?

Good luck on your exam! Remember: if you understand how amino acid properties drive folding and how folding determines function, you have the conceptual framework to answer virtually any AP question about proteins. 🧬